

2016.0 RANGE ROVER (LG), 100-00

# GENERAL INFORMATION

DESCRIPTION AND OPERATION

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## TRANSFER CASE CONTROL MODULE (TCCM)

### CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

## NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Transfer Case Control Module (TCCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Transfer Case](#) (308-07C Transfer Case -

## Vehicles With: Twin Speed Transfer Case, Diagnosis and Testing).

| DTC      | DESCRIPTION                                                                      | POSSIBLE CAUSES                                                                                                                                                                                                                       | ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| P0712-00 | Transmission Fluid Temperature Sensor 'A' Circuit Low - No sub type information  | <p><b>NOTE:</b></p> <div>Circuit reference TEMP</div> <ul style="list-style-type: none"> <li>Transfer case motor temperature sensor circuit short circuit to ground</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTE:</b></p> <div>Transfer case motor temperature sensor (temperature signal circuit to motor position sensor ground circuit) resistance should be 1.0k ohms to 30.0k ohms</div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case motor temperature sensor circuit for short circuit to ground</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| P0713-00 | Transmission Fluid Temperature Sensor 'A' Circuit High - No sub type information | <p><b>NOTE:</b></p> <div>Circuit reference TEMP</div> <ul style="list-style-type: none"> <li>Transfer case motor temperature sensor circuit short circuit to power</li> <li>Transfer case control module internal failure</li> </ul>  | <p><b>NOTE:</b></p> <div>Transfer case motor temperature sensor (temperature signal circuit to motor position sensor ground circuit) resistance should be 1.0k ohms to 30.0k ohms.</div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case motor temperature sensor circuit for short circuit to power</li> <li>Using the manufacturer</li> </ul>                                                                                                                                                                          |

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|          |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| P080A-54 | Clutch Position Not Learned - Missing calibration | <p><b>NOTE:</b></p> <div> <p>Circuit reference HALL V+ / HALL INA / HALL INB / HALL GND / SOLENOID+ / SOLENOID-</p> </div> <ul style="list-style-type: none"> <li>Transfer case clutch not calibrated</li> <li>Transfer case motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Transfer case solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Transfer case fluid contaminated with water</li> <li>Transfer case clutch worn/burnt</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTES:</b></p> <div> <ul style="list-style-type: none"> <li>If additional transfer case circuit fault related DTCs are also set, perform the relevant corrective action(s) first and retest.</li> <li>Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.</li> <li>Transfer case solenoid resistance should be 0.2 ohms to 10.0 ohms.</li> </ul> </div> <ul style="list-style-type: none"> <li>Using the manufacturer approved diagnostic system, perform routine - Transfer Case Replacement</li> <li>Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul> |

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|          |                                                                    |                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check the transfer case solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Refer to the relevant section of the workshop manual and check the transfer case fluid level/condition. If water is present in the fluid, check the integrity of the transfer case breather and install a new transfer case and perform routine - Transfer Case Replacement</li> <li>■ Check the integrity of the driveshafts, differentials and half shafts. Rectify as necessary and install a new transfer case and perform routine - Transfer Case Replacement</li> <li>■ Using the manufacturer approved diagnostic system, perform routine - Transfer Case Replacement. Clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| P181E-64 | Clutch Actuator Range/Performance<br>- Signal plausibility failure | <p><b>NOTE:</b></p> <div> <p>Circuit reference<br/>MOTOR+ /<br/>MOTOR- /<br/>HALL V+ /<br/>HALL INA /<br/>HALL INB /<br/>HALL GND /<br/>SOLENOID+ /<br/>SOLENOID-</p> </div> <ul style="list-style-type: none"> <li>■ Transfer case motor circuit short circuit to ground, short circuit</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

to power, open circuit, high resistance

- Transfer case motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
- Transfer case solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance
- Transfer case motor failure
- Transfer case solenoid failure
- Transfer case control module internal failure
- Transfer case internal failure

#### NOTES:

- If additional transfer case circuit fault related DTCs are also set, perform the relevant corrective action(s) first and retest.
- Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.
- Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.
- Transfer case solenoid resistance should be 0.2 ohms to 10.0 ohms.

- Refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
- Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
- Refer to the electrical circuit diagrams and check the

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|          |                                                     |                                                                                                                                                                                                                                             | <p>transfer case solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case motor and perform routine - Transfer Case Replacement</li> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case solenoid and perform routine - Transfer Case Replacement</li> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case and perform routine - Transfer Case Replacement</li> </ul> |
| P181E-93 | Clutch Actuator Range/Performance<br>- No operation | <p><b>NOTE:</b></p> <div> <p>Circuit reference HALL V+ / HALL INA / HALL INB / HALL GND / SOLENOID+ / SOLENOID-</p> </div> <ul style="list-style-type: none"> <li>■ Transfer case motor position sensor circuit short circuit to</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

ground, short circuit to power, open circuit, high resistance

- Transfer case solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance
- Transfer case motor failure
- Transfer case solenoid failure
- Transfer case control module internal failure
- Transfer case internal failure

#### NOTES:

- If additional transfer case circuit fault related DTCs are also set, perform the relevant corrective action(s) first and retest.
  - Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.
  - Transfer case solenoid resistance should be 0.2 ohms to 10.0 ohms.
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- Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
  - Refer to the electrical circuit diagrams and check the transfer case solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance
  - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case motor and perform routine -



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|          |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Transfer Case Replacement</p> <ul style="list-style-type: none"> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case solenoid and perform routine - Transfer Case Replacement</li> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case and perform routine - Transfer Case Replacement</li> </ul> |
| P181F-64 | Clutch Control System Performance - Signal plausibility failure | <p><b>NOTE:</b></p> <div> <p>Circuit reference<br/> MOTOR+ /<br/> MOTOR- /<br/> HALL V+ /<br/> HALL INA /<br/> HALL INB /<br/> HALL GND /<br/> SOLENOID+ /<br/> SOLENOID-</p> </div> <ul style="list-style-type: none"> <li>■ Transfer case motor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Transfer case motor position sensor circuit short circuit to ground, short circuit</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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|  |  |  | <p>to power, open circuit, high resistance</p> <ul style="list-style-type: none"><li>▪ Transfer case solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li><li>▪ Transfer case control module internal failure</li><li>▪ Transfer case motor failure</li><li>▪ Transfer case solenoid failure</li><li>▪ Transfer case internal failure</li></ul> | <p><b>NOTES:</b></p> <div><ul style="list-style-type: none"><li>▪ If additional transfer case circuit fault related DTCs are also set, perform the relevant corrective action(s) first and retest.</li><li>▪ Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.</li><li>▪ Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.</li><li>▪ Transfer case solenoid resistance should be 0.2 ohms to 10.0 ohms.</li></ul></div> <ul style="list-style-type: none"><li>▪ Refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li><li>▪ Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li><li>▪ Refer to the electrical circuit diagrams and check the</li></ul> |
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|          |                                                         |                                                                                                                                                                                                                                                                                                               | <p>transfer case solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case motor and perform routine - Transfer Case Replacement</li> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case solenoid and perform routine - Transfer Case Replacement</li> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case and perform routine - Transfer Case Replacement</li> </ul> |
| P183A-54 | Range Change Mechanism Failure<br>- Missing calibration | <p><b>NOTE:</b></p> <div>           Circuit reference<br/>           MOTOR+ /<br/>           MOTOR- /<br/>           HALL V+ /<br/>           HALL INA /<br/>           HALL INB /<br/>           HALL GND /<br/>           SENSOR PWR /<br/>           SENSOR SIG /<br/>           SENSOR GND         </div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

- Transfer case selector fork position sensor not calibrated
- Transfer case motor circuit short circuit to ground, short circuit to power, open circuit, high resistance
- Transfer case motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
- Transfer case selector fork position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
- Transfer case motor failure
- Transfer case selector fork position sensor failure
- Transfer case control module internal failure
- Transfer case internal failure

#### NOTES:

- If additional transfer case circuit fault related DTCs are also set, perform the relevant corrective action(s) first and retest.
  - Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.
  - Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.
  - Transfer case selector fork position sensor voltage (supply circuit to ground circuit) should be 4.0V to 6.0V.
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- Using the manufacturer approved diagnostic system, perform routine - Transfer Case Replacement. Clear the DTCs and retest
  - Refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
  - Refer to the electrical circuit

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|          |                                                            |  | <p>diagrams and check the transfer case motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check the transfer case selector fork position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case motor and perform routine - Transfer Case Replacement</li> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case selector fork position sensor and perform routine - Transfer Case Replacement</li> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case and perform routine - Transfer Case Replacement</li> </ul> |  |
| P183A-62 | Range Change Mechanism Failure<br>- Signal compare failure |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |

|  |  |  |  | <b>NOTE:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>NOTES:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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|  |  |  |  | <div>Circuit<br/>reference HALL<br/>V+ / HALL INA<br/>/ HALL INB /<br/>HALL GND /<br/>SENSOR PWR /<br/>SENSOR SIG /<br/>SENSOR GND</div> <ul style="list-style-type: none"><li>▪ Transfer case motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li><li>▪ Transfer case selector fork position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li><li>▪ Transfer case selector fork position sensor failure</li><li>▪ Transfer case motor failure</li><li>▪ Transfer case control module internal failure</li></ul> | <div><ul style="list-style-type: none"><li>▪ If additional transfer case circuit fault related DTCs are also set, perform the relevant corrective action(s) first and retest.</li><li>▪ Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.</li><li>▪ Transfer case selector fork position sensor voltage (supply circuit to ground circuit) should be 4.0V to 6.0V.</li></ul></div> <ul style="list-style-type: none"><li>▪ Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li><li>▪ Refer to the electrical circuit diagrams and check the transfer case selector fork position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li><li>▪ Using the manufacturer approved diagnostic</li></ul> |

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|          |                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>system, clear the DTCs and retest. If the fault persists, install a new transfer case selector fork position sensor and perform routine - Transfer Case Replacement</p> <ul style="list-style-type: none"> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case motor and perform routine - Transfer Case Replacement</li> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| P183A-67 | <p>Range Change Mechanism Failure</p> <p>- Signal incorrect after event</p> | <p><b>NOTE:</b></p> <div> <p>Circuit reference<br/>MOTOR+ /<br/>MOTOR- /<br/>HALL V+ /<br/>HALL INA /<br/>HALL INB /<br/>HALL GND</p> </div> <ul style="list-style-type: none"> <li>■ Transfer case motor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Transfer case motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Transfer case motor failure</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

- Transfer case control module internal failure
- Transfer case internal failure
- Transmission fault - High drag torque in Neutral

#### NOTES:

- If additional transfer case circuit fault related DTCs are also set, perform the relevant corrective action(s) first and retest.
  - If any transmission control module DTCs are also set, perform the relevant corrective action(s) first and retest.
  - Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.
  - Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.
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- Refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
  - Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance



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|          |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case motor and perform routine - Transfer Case Replacement</li> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case and perform routine - Transfer Case Replacement</li> <li>▪ Refer to the relevant section of the workshop manual and test the transmission</li> </ul> |
| P183A-77 | Range Change Mechanism Failure<br>- Commanded position not reachable | <p><b>NOTE:</b></p> <div> <p>Circuit reference<br/>MOTOR+ /<br/>MOTOR- /<br/>HALL V+ /<br/>HALL INA /<br/>HALL INB /<br/>HALL GND</p> </div> <ul style="list-style-type: none"> <li>▪ Transfer case motor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Transfer case motor position sensor circuit short circuit to ground, short circuit to power, open</li> </ul> | <p><b>NOTES:</b></p> <div> <ul style="list-style-type: none"> <li>▪ Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.</li> <li>▪ Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.</li> </ul> </div>                                                                                                                                                                                                                                                                                               |

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|          |                                                             | <p>circuit, high resistance</p> <ul style="list-style-type: none"> <li>Transfer case control module internal failure</li> <li>Transfer case internal failure</li> </ul>                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case and perform routine - Transfer Case Replacement</li> </ul> |
| P183A-93 | <p>Range Change Mechanism Failure</p> <p>- No operation</p> | <p><b>NOTE:</b></p> <div> <p>Circuit reference<br/> MOTOR+ /<br/> MOTOR- /<br/> HALL V+ /<br/> HALL INA /<br/> HALL INB /<br/> HALL GND</p> </div> <ul style="list-style-type: none"> <li>Transfer case motor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Transfer case motor position sensor</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

circuit short circuit to ground, short circuit to power, open circuit, high resistance

- Transfer case motor failure
- Transfer case control module internal failure
- Transfer case internal failure
- Transmission fault - High drag torque in Neutral

#### NOTES:

- If additional transfer case circuit fault related DTCs are also set, perform the relevant corrective action(s) first and retest.
- If any transmission control module DTCs are also set, perform the relevant corrective action(s) first and retest.
- Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.
- Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.

- Refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
- Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance

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|          |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case motor and perform routine - Transfer Case Replacement</li> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case and perform routine - Transfer Case Replacement</li> <li>▪ Refer to the relevant section of the workshop manual and test the transmission</li> </ul> |
| P186D-71 | Clutch Actuator Stuck - Actuator stuck | <p><b>NOTE:</b></p> <div> <p>Circuit reference<br/> MOTOR+ /<br/> MOTOR- /<br/> HALL V+ /<br/> HALL INA /<br/> HALL INB /<br/> HALL GND /<br/> SOLENOID+ /<br/> SOLENOID-</p> </div> <ul style="list-style-type: none"> <li>▪ Transfer case motor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Transfer case motor position sensor circuit short circuit to</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

ground, short circuit to power, open circuit, high resistance

- Transfer case solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance
- Transfer case motor failure
- Transfer case control module internal failure
- Transfer case internal failure

#### NOTES:

- If additional transfer case circuit fault related DTCs are also set, perform the relevant corrective action(s) first and retest.
- Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.
- Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.
- Transfer case solenoid resistance should be 0.2 ohms to 10.0 ohms.

- Refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
- Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
- Refer to the electrical circuit diagrams and check the

|          |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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|          |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                           | <p>transfer case solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case motor and perform routine - Transfer Case Replacement</li> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> <li>■ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case and perform routine - Transfer Case Replacement</li> </ul> |
| P18A0-64 | Transfer Case Motion/Position Sensor - Signal plausibility failure | <p><b>NOTE:</b></p> <div> <p>Circuit reference<br/>MOTOR+ /<br/>MOTOR- /<br/>HALL V+ /<br/>HALL INA /<br/>HALL INB /<br/>HALL GND</p> </div> <ul style="list-style-type: none"> <li>■ Transfer case motor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Transfer case motor position sensor circuit short circuit to ground, short circuit</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

to power, open circuit, high resistance

- Transfer case motor failure
- Transfer case control module internal failure

#### NOTES:

- If additional transfer case circuit fault related DTCs are also set, perform the relevant corrective action(s) first and retest.
  - Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.
  - Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.
- 
- Refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
  - Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
  - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case motor and perform routine -

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|          |                                                                        |                                                                                                                                                                                                                                                                                        | <p>Transfer Case Replacement</p> <ul style="list-style-type: none"> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul>                                                                                                                                                                                                                                                              |
| P18A1-11 | Transfer Case Range Position Sensor Circuit - Circuit short to ground  | <p><b>NOTE:</b></p> <div>Circuit reference<br/>SENSOR PWR /<br/>SENSOR SIG /<br/>SENSOR GND</div> <ul style="list-style-type: none"> <li>Transfer case selector fork position sensor circuit short circuit to ground</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTE:</b></p> <div>Transfer case selector fork position sensor voltage (supply circuit to ground circuit) should be 4.0V to 6.0V.</div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case selector fork position sensor circuit for short circuit to ground</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| P18A1-12 | Transfer Case Range Position Sensor Circuit - Circuit short to battery | <p><b>NOTE:</b></p> <div>Circuit reference<br/>SENSOR PWR /<br/>SENSOR SIG /<br/>SENSOR GND</div> <ul style="list-style-type: none"> <li>Transfer case selector fork position sensor circuit short circuit to power</li> <li>Transfer case control</li> </ul>                          | <p><b>NOTE:</b></p> <div>Transfer case selector fork position sensor voltage (supply circuit to ground circuit) should be 4.0V to 6.0V.</div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case selector fork position sensor circuit for short circuit to power</li> </ul>                                                                                                                                                                                                           |



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|          |                                                                               | module internal failure                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul>                                                                                                                                                                                                                                                                                               |
| P18A1-16 | Transfer Case Range Position Sensor Circuit - Circuit voltage below threshold | <p><b>NOTE:</b></p> <div>Circuit reference<br/>SENSOR PWR /<br/>SENSOR SIG /<br/>SENSOR GND</div> <ul style="list-style-type: none"> <li>Transfer case selector fork position sensor circuit short circuit to ground</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTE:</b></p> <div>Transfer case selector fork position sensor voltage (supply circuit to ground circuit) should be 4.0V to 6.0V.</div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case selector fork position sensor circuit for short circuit to ground</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| P18A1-17 | Transfer Case Range Position Sensor Circuit - Circuit voltage above threshold | <p><b>NOTE:</b></p> <div>Circuit reference<br/>SENSOR PWR /<br/>SENSOR SIG /<br/>SENSOR GND</div> <ul style="list-style-type: none"> <li>Transfer case selector fork position sensor circuit short circuit to power</li> <li>Transfer case control module internal</li> </ul>          | <p><b>NOTE:</b></p> <div>Transfer case selector fork position sensor voltage (supply circuit to ground circuit) should be 4.0V to 6.0V.</div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case selector fork position sensor circuit for short circuit to power</li> <li>Using the manufacturer</li> </ul>                                                                                                                                                                           |

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|          |                                                                 | failure                                                                                                                                                                                                                                                                                      | approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement                                                                                                                                                                                                                                                                                                                                                        |
| P18A2-11 | Transfer Case Actuator Motor Circuit - Circuit short to ground  | <p><b>NOTE:</b></p> <div>Circuit reference<br/>MOTOR+ /<br/>MOTOR-</div> <ul style="list-style-type: none"> <li>Transfer case motor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTE:</b></p> <div>Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.</div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| P18A2-12 | Transfer Case Actuator Motor Circuit - Circuit short to battery | <p><b>NOTE:</b></p> <div>Circuit reference<br/>MOTOR+ /<br/>MOTOR-</div> <ul style="list-style-type: none"> <li>Transfer case motor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Transfer case control module internal</li> </ul>         | <p><b>NOTE:</b></p> <div>Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.</div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the manufacturer approved diagnostic</li> </ul>                                                                                                                                                      |

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|          |                                                                  | failure                                                                                                                                                                                                                                                                                      | system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement                                                                                                                                                                                                                                                                                                                                                                            |
| P18A2-13 | Transfer Case Actuator Motor Circuit - Circuit open              | <p><b>NOTE:</b></p> <div>Circuit reference<br/>MOTOR+ /<br/>MOTOR-</div> <ul style="list-style-type: none"> <li>Transfer case motor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTE:</b></p> <div>Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.</div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| P18A3-16 | Transfer Case Solenoid Circuit - Circuit voltage below threshold | <p><b>NOTE:</b></p> <div>Circuit reference<br/>SOLENOID+ /<br/>SOLENOID-</div> <ul style="list-style-type: none"> <li>Transfer case solenoid circuit short circuit to ground</li> <li>Transfer case control module internal failure</li> </ul>                                               | <p><b>NOTE:</b></p> <div>Transfer case solenoid resistance should be 0.2 ohms to 10.0 ohms.</div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case solenoid circuit for short circuit to ground</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case</li> </ul>                                                                                                                 |

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|          |                                                                  |                                                                                                                                                                                                                                                                       | control module and perform routine - Transfer Case Replacement                                                                                                                                                                                                                                                                                                                                                                                                                              |
| P18A3-17 | Transfer Case Solenoid Circuit - Circuit voltage above threshold | <p><b>NOTE:</b></p> <div>Circuit reference SOLENOID+ / SOLENOID-</div> <ul style="list-style-type: none"> <li>Transfer case solenoid circuit short circuit to power</li> <li>Transfer case control module internal failure</li> </ul>                                 | <p><b>NOTE:</b></p> <div>Transfer case solenoid resistance should be 0.2 ohms to 10.0 ohms.</div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case solenoid circuit for short circuit to power</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul>               |
| P18A4-12 | Motion/Position Sensor Power Circuit - Circuit short to battery  | <p><b>NOTE:</b></p> <div>Circuit reference HALL V+ / HALL INA / HALL INB / HALL GND</div> <ul style="list-style-type: none"> <li>Transfer case motor position sensor circuit short circuit to power</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTE:</b></p> <div>Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.</div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to power</li> </ul> |

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|          |                                                                        |                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| P18A4-13 | Motion/Position Sensor Power Circuit - Circuit open                    | <p><b>NOTE:</b></p> <div> <p>Circuit reference HALL V+ / HALL INA / HALL INB / HALL GND</p> </div> <ul style="list-style-type: none"> <li>Transfer case motor position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTE:</b></p> <div> <p>Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.</p> </div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| P18A4-16 | Motion/Position Sensor Power Circuit - Circuit voltage below threshold |                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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|          |                                                                     | <p><b>NOTE:</b></p> <div> <p>Circuit reference HALL V+ / HALL INA / HALL INB / HALL GND</p> </div> <ul style="list-style-type: none"> <li>Transfer case motor position sensor circuit short circuit to ground</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTE:</b></p> <div> <p>Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.</p> </div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| P18A5-11 | Motion Position Sensor "A" Signal Circuit - Circuit short to ground | <p><b>NOTE:</b></p> <div> <p>Circuit reference HALL INA</p> </div> <ul style="list-style-type: none"> <li>Transfer case motor position sensor circuit short circuit to ground</li> <li>Transfer case control module internal failure</li> </ul>                                 | <p><b>NOTE:</b></p> <div> <p>Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.</p> </div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit</li> </ul>                                                                                                                                                                                                                                                                                                         |

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|          |                                                                              |                                                                                                                                                                                                                                                                                                               | <p>diagrams and check the transfer case motor position sensor circuit for short circuit to ground</p> <ul style="list-style-type: none"> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| P18A5-15 | Motion Position Sensor "A" Signal Circuit - Circuit short to battery or open | <p><b>NOTE:</b></p> <div style="border: 1px solid black; padding: 5px; margin: 10px 0;"> <p>Circuit reference HALL INA</p> </div> <ul style="list-style-type: none"> <li>Transfer case motor position sensor circuit short circuit to power</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTE:</b></p> <div style="border: 1px solid black; padding: 5px; margin: 10px 0;"> <p>Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.</p> </div> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to power</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| P18A6-11 | Motion Position Sensor "B" Signal Circuit - Circuit short to ground          |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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|          |                                                                              | <p><b>NOTE:</b></p> <div>Circuit reference HALL INB</div> <ul style="list-style-type: none"> <li>Transfer case motor position sensor circuit short circuit to ground</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTE:</b></p> <div> <p>Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.</p> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the transfer case motor position sensor circuit for short circuit to ground</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> </div> |
| P18A6-15 | Motion Position Sensor "B" Signal Circuit - Circuit short to battery or open | <p><b>NOTE:</b></p> <div>Circuit reference HALL INB</div> <ul style="list-style-type: none"> <li>Transfer case motor position sensor circuit short circuit to ground</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTE:</b></p> <div> <p>Transfer case motor position sensor voltage (sensor supply to sensor ground) should be 8V to 11V, resistance (signal A circuit to ground circuit) should be 16.0k ohms to 20.0k ohms and resistance (signal B circuit to ground circuit) should be 16.0k ohms to 20.0k ohms.</p> <ul style="list-style-type: none"> <li>Refer to the electrical circuit</li> </ul> </div>                                                                                                                                                                                                                                                                                                         |



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|          |                                                 |                                                                                                                                                                                                                                                                                                                                                                                      | <p>diagrams and check the transfer case motor position sensor circuit for short circuit to ground</p> <ul style="list-style-type: none"> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| P18A7-24 | Range Change Switch Circuit - Signal stuck high | <p><b>NOTE:</b></p> <div style="border: 1px solid black; padding: 5px; margin: 10px 0;"> <p>Circuit reference SW HI/LO RATIO</p> </div> <ul style="list-style-type: none"> <li>Transfer case high/low range switch stuck active</li> <li>Transfer case high/low range switch signal circuit short circuit to power</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTES:</b></p> <div style="border: 1px solid black; padding: 5px; margin: 10px 0;"> <ul style="list-style-type: none"> <li>This DTC will be set if the switch signal is active for &gt; 255 seconds.</li> <li>This DTC may be induced by the driver.</li> </ul> </div> <ul style="list-style-type: none"> <li>Check the operation of the transfer case high/low range switch</li> <li>Using the manufacturer approved diagnostic system, check datalogger signal - TCCM Input Status (0x1983). Refer to the electrical circuit diagrams and check the transfer case high/low range switch signal circuits for short circuit to power</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| P2785-98 | Clutch Actuator Temperature Too                 | <ul style="list-style-type: none"> <li>Transfer case motor temperature too</li> </ul>                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

High - Component or system over temperature

high (> 150°C)

- Transfer case motor circuit short circuit to ground, short circuit to power, open circuit, high resistance
- Transfer case control module internal failure

**NOTES:**

- Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.
- This DTC may be induced by prolonged off-road driving.

- Allow the vehicle to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest
- Refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
- Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement

P2787-98

Clutch Temperature Too High - Component or system over temperature

- Transfer case clutch temperature too high
- Incorrect specification wheels/tires installed - Mismatch between axle speeds
- Front/rear driveshaft failure
- Front/rear differential failure
- Front/rear left/right halfshaft failure

**NOTE:**

This DTC may be induced by prolonged off-road driving.

- Allow the vehicle to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest
- Check the wheel and tire specifications
- Check the integrity of the

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|          |                                                                            |                                                                                                                                                                          | <p>front and rear driveshafts</p> <ul style="list-style-type: none"> <li>▪ Check the integrity of the front and rear differentials</li> <li>▪ Check the integrity of the halfshafts</li> </ul>                                                                                                                                                             |
| U0001-81 | High Speed CAN Communication Bus - Invalid serial data received            | <ul style="list-style-type: none"> <li>▪ Invalid data received from another control module via the high speed CAN bus (powertrain)</li> </ul>                            | <ul style="list-style-type: none"> <li>▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index</li> </ul>                                                                                 |
| U0001-87 | High Speed CAN Communication Bus - Missing message                         | <ul style="list-style-type: none"> <li>▪ Missing message from another control module via the high speed CAN bus (powertrain)</li> </ul>                                  | <ul style="list-style-type: none"> <li>▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index</li> </ul>                                                                              |
| U0001-88 | High Speed CAN Communication Bus - Bus off                                 | <ul style="list-style-type: none"> <li>▪ High speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>            | <ul style="list-style-type: none"> <li>▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                          |
| U0300-00 | Internal Control Module Software Incompatibility - No sub type information | <ul style="list-style-type: none"> <li>▪ Transfer case control module is not configured correctly</li> <li>▪ Incorrect transfer case control module installed</li> </ul> | <ul style="list-style-type: none"> <li>▪ Using the manufacturer approved diagnostic system, re-configure the transfer case control module with the latest level software</li> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform</li> </ul> |

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|          |                                                          |                                                                                                                                                                                                                                    | routine - Transfer Case Replacement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| U3000-1F | Control Module - Circuit intermittent                    | <ul style="list-style-type: none"> <li>Transfer case control module power or ground circuit open circuit, high resistance</li> <li>Battery/charging system fault</li> <li>Transfer case control module internal failure</li> </ul> | <ul style="list-style-type: none"> <li>Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance</li> <li>Refer to the relevant section of the workshop manual and test the battery and charging system</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| U3000-47 | Control Module - Watchdog/safety MicroController failure | <ul style="list-style-type: none"> <li>Transfer case control module power or ground circuit open circuit, high resistance</li> <li>Battery/charging system fault</li> </ul>                                                        | <p><b>NOTE:</b></p> <div style="border: 1px solid black; padding: 10px; margin: 10px 0;"> <p>This DTC may be set during a forced reset by the manufacturer approved diagnostic system.</p> </div> <ul style="list-style-type: none"> <li>Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance</li> <li>Refer to the relevant section</li> </ul>                                                                        |

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|          |                                              |                                                                                                                                                                                                                                                                                                                         | of the workshop manual and test the battery and charging system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| U3000-49 | Control Module - Internal electronic failure | <ul style="list-style-type: none"> <li>Transfer case control module internal failure</li> </ul>                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| U3000-4B | Control Module - Over temperature            | <p><b>NOTE:</b></p> <div style="border: 1px solid black; padding: 10px; margin: 10px 0;"> <p>Circuit reference<br/>SOLENOID+ /<br/>SOLENOID-</p> </div> <ul style="list-style-type: none"> <li>Transfer case solenoid circuit short circuit to ground</li> <li>Transfer case control module internal failure</li> </ul> | <p><b>NOTE:</b></p> <div style="border: 1px solid black; padding: 10px; margin: 10px 0;"> <p>Transfer case solenoid resistance should be 0.2 ohms to 10.0 ohms.</p> </div> <ul style="list-style-type: none"> <li>Allow the vehicle to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, refer to the electrical circuit diagrams and check the transfer case solenoid circuit for short circuit to ground</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| U3000-55 | Control Module - Not configured              | <ul style="list-style-type: none"> <li>Car configuration file mismatch with vehicle specification</li> <li>Incorrect transfer case control module installed</li> </ul>                                                                                                                                                  | <ul style="list-style-type: none"> <li>Using the manufacturer approved diagnostic system, check and up-date the car configuration file as necessary</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists,</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                            |

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|          |                                                       |                                                                                                                                                                                                                                                                                                                                                                           | install a new transfer case control module and perform routine - Transfer Case Replacement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| U3000-98 | Control Module - Component or system over temperature | <p><b>NOTE:</b></p> <div style="border: 1px solid black; padding: 10px; margin: 10px 0;"> <p>Circuit reference<br/>MOTOR+ /<br/>MOTOR-</p> </div> <ul style="list-style-type: none"> <li>▪ Transfer case motor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Transfer case control module internal failure</li> </ul> | <p><b>NOTES:</b></p> <div style="border: 1px solid black; padding: 10px; margin: 10px 0;"> <ul style="list-style-type: none"> <li>▪ Transfer case motor resistance should be 0.2 ohms to 10.0 ohms.</li> <li>▪ This DTC may be induced by excessive high/low range changes.</li> <li>▪ This DTC may be induced by prolonged off-road driving.</li> </ul> </div> <ul style="list-style-type: none"> <li>▪ Allow the vehicle to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, refer to the electrical circuit diagrams and check the transfer case motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul> |
| U3003-16 | Battery voltage - Circuit voltage below threshold     | <ul style="list-style-type: none"> <li>▪ Transfer case control module power or ground circuit open circuit, high resistance</li> <li>▪ Battery/charging system fault</li> </ul>                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the transfer case control module</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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|          |                                                   | <ul style="list-style-type: none"> <li>Transfer case control module internal failure</li> </ul>                                                                                                                                    | <p>power and ground circuits for open circuit, high resistance</p> <ul style="list-style-type: none"> <li>Refer to the relevant section of the workshop manual and test the battery and charging system</li> <li>Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul>   |
| U3003-17 | Battery voltage - Circuit voltage above threshold | <ul style="list-style-type: none"> <li>Battery/charging system fault</li> </ul>                                                                                                                                                    | <p><b>NOTE:</b></p> <div style="border: 1px solid black; padding: 10px; margin: 10px 0;"> <p>This DTC may be set when the vehicle is jump started.</p> </div> <ul style="list-style-type: none"> <li>Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the relevant section of the workshop manual and test the battery and charging system</li> </ul> |
| U3003-1C | Battery voltage - Circuit voltage out of range    | <ul style="list-style-type: none"> <li>Transfer case control module power or ground circuit open circuit, high resistance</li> <li>Battery/charging system fault</li> <li>Transfer case control module internal failure</li> </ul> | <ul style="list-style-type: none"> <li>Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111). Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance</li> <li>Refer to the relevant section of the workshop manual and test the battery and charging system</li> </ul>      |

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|          |                                          |                                                                                                                                                                     | <ul style="list-style-type: none"> <li>▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transfer case control module and perform routine - Transfer Case Replacement</li> </ul>                                                                                                     |
| U3003-62 | Battery Voltage - Signal compare failure | <ul style="list-style-type: none"> <li>▪ Mismatch between the voltage at the transfer case control module and the voltage value broadcast on the CAN bus</li> </ul> | <ul style="list-style-type: none"> <li>▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD111) - and compare it to battery voltage. Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance</li> </ul> |